

Jinha Choi

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RESEARCH INTERESTS

My research focuses on dexterous manipulation with high-DoF robotic hands, incorporating physics- and control-informed priors to develop generalizable robot intelligence for unstructured environments and real-world deployment.

EDUCATION

Yonsei University

Mar. 2021 – Present

B.S. Candidate in Computer Science

B.S. Candidate in Nano Science and Engineering

GPA: 3.89 / 4.3

West Vancouver Secondary School, Canada

Sep. 2014 – Jun. 2020

International Baccalaureate Diploma

RESEARCH

Yonsei RoPhi Lab

Sep. 2026 – Present

Advised by Prof. Yonghyeon Lee

Developing high-DoF dexterous manipulation methods with physics-informed priors to improve real-world adaptability and generalization in unstructured environments.

KAIST HuGeLab

Jun. 2026 – Aug. 2026

Summer Internship, Advised by Prof. Beomjoon Kim

Developed and deployed learning-based locomotion policies for the Unitree G1 and a custom 108 kg humanoid, with an emphasis on robustness across diverse environments and real-world conditions.

Yonsei DILLAB

Jan. 2026 – Jun. 2026

Advised by Prof. Jongmin Lee

Developed an offline-RL method that distills a value-aware multi-step flow policy into an efficient one-step policy through state-wise entropic optimal transport, improving multimodal policy learning across benchmark tasks.

Yonsei RLLAB

Jul. 2025 – Dec. 2025

Advised by Prof. Youngwoon Lee

Built humanoid loco-manipulation baselines using motion priors from retargeted human motion. Analyzed how retargeting choices affect motion quality and stability, and studied reusable motion components with less task-specific reward engineering.

Yonsei MIRLAB

Sep. 2024 – Mar. 2025

Advised by Prof. Youngjae Yu

Studied multimodal learning and Isaac Sim workflows for robot learning. Contributed to an LLM scientific-verification benchmark research through dataset processing, labeling, and taxonomy design.

PUBLICATIONS

- (2) **J. Choi***, J. Shon*, J. Jeon, and J. Lee, “Learning Multimodal One-step Flow Policy via Value-weighted Optimal Transport,” (2026). Manuscript submitted to NeurIPS 2026.

*Equal contribution.

- (1) G. Son, J. Hong, H. Fan, H. Nam, H. Ko, S. Lim, J. Song, **J. Choi**, G. Paulo, Y. Yu, and S. Biderman, “When AI Co-Scientists Fail: SPOT—a Benchmark for Automated Verification of Scientific Research,” (2025). arXiv:2505.11855.

EXPERIENCE & PROJECTS

Humanoid Deployment – Locomotion

Jul. 2026

Advised by Prof. Beomjoon Kim

Deployed trained locomotion policies on Unitree G1 and a custom 108 kg humanoid through simulation training, sim-to-sim validation, and sim-to-real evaluation.

Franka Panda Teleoperation Data Collection*Jul. 2026**Advised by Prof. Beomjoon Kim*

Integrated a Franka Panda with a 3D mouse, Touch haptic device, and Meta Quest for teleoperated demonstration collection.

Humanoid Fixed-Base Manipulation*Mar. 2026 – Jun. 2026**Advised by Prof. Youngwoon Lee*

Investigated PPO exploration limits for humanoid fixed-base manipulation in massively parallel simulation. Evaluated PPO's variants, including SAPG and EPO, against standard PPO and analyzed methods for improving exploration.

A1-Challenge Autonomous Driving Competition*Mar. 2026 – May 2026**Qualified in Nov. 2025; 4th Place; hosted by the Ministry of Trade and KIAP*

Developed an end-to-end racing policy in the MORAI simulator. Built behavioral-cloning and DAGger baselines using targeted corner-case data, then explored reinforcement learning to improve robustness and driving performance.

Physical AI Robotics Hackathon - Single Arm Manipulation*Mar. 2026**Hosted by Kernel Academy and Doosan Robotics*

Built a speech-guided pick-and-place pipeline using a Doosan E0509 arm, RGB-D sensing, Whisper, SAM2, GPT-4o-based reasoning, and ROS2. Added calibrated 3D localization, task verification, and error-recovery logic.

Language-Conditioned Image Inpainting with LLMs and Diffusion Models*Jul. 2024 – Aug. 2024**OUTTA AI Bootcamp Project*

Developed a language-conditioned image editing pipeline using a pretrained LLM to extract clothing attributes, semantic segmentation to localize garments, and diffusion-based inpainting to synthesize the specified appearance.

Korea Defense Intelligence Command – Translator*Mar. 2023 – Sep. 2024**Military Service*

Translated classified documents and interpreted classified meetings for KDIC and partner agencies, including the National Intelligence Service and the U.S. National Geospatial-Intelligence Agency.

VEX Robotics*Sep. 2017 – Jun. 2020**Team Lead, 1010B, WVSS Robotics*

Led a 12-member team through strategy, mechanical and electrical design, programming, autonomous control, and competitive driving across three VEX Robotics seasons, including the World Championship.

AWARDS & HONORS

Hyundai NGV Robotics Track H-Mobility Class Certificate*Mar. 2026 – Jul. 2026*

Completed Hyundai's Robotics Track, a program on robotics and robot learning, and received H-Mobility certificate from Hyundai.

2nd Place, Physical AI Robotics Hackathon*Mar. 2026*

Built a speech-guided pick-and-place pipeline using a Doosan E0509 arm, RGB-D sensing, Whisper, SAM2, GPT-4o-based reasoning, and ROS2. Added calibrated 3D localization, task verification, and error-recovery logic

Honor Roll, Yonsei University*Jul. 2022***1st Place, Yonsei-Nexon RC Creative Platform***Mar. 2021 – Dec. 2021*

Led a five-member team to design and prototype an automated revolving-door assistance system for people with mobility impairments; ranked 1st of 90 teams and received a \$10,000 award.

Division Finalist, VEX Robotics World Championship – Kentucky, USA*Nov. 2019*

Reached the Science Division final at the VEX Robotics World Championship, finishing among the top teams in a 100-team division within a field of total 600 qualified teams.

1st Place, VEX Robotics Canada BC Provincial Championship*Jun. 2019*

Led Team 1010B to win the British Columbia Provincial Championship and qualify for the VEX Robotics World Championship.